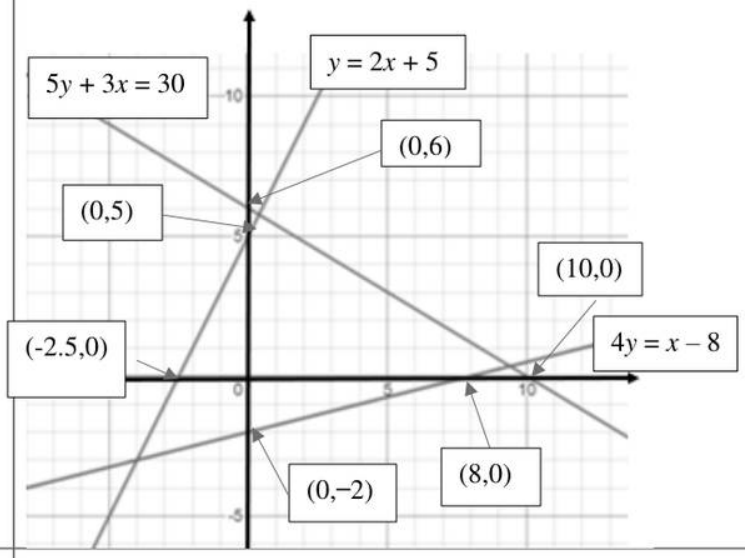
Type MS · Shading or describing a region

Answer 1 of 11 · Question 5 part b · 1 mark

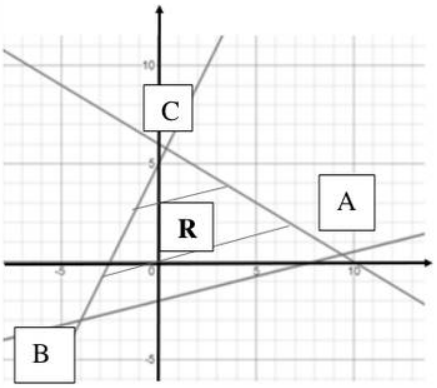
2023 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

5 (a)



B3

(b)



B1

(c)

	A	B	C
x	9.4	−4	0.4
y	0.4	−3	5.8
P	$2\times9.4-5\times0.4$ $=16.8$	$2\times-4-5\times-3$ $=7$	$2\times0.4-5\times5.8$ $=-28.2$ [Exact value is] $\left[-\frac{365}{13}\approx-28.1\right]$

M1

M1A1

Allow ± 0.1 on the coordinates in each case.

Total 7 marks

Type MS · Shading or describing a region

Answer 1 of 11 · Question 5 part b · 1 mark

2023 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
The tolerance for all marks in this question is ± half a small square.		
(a)	**B1**	For one of the lines correctly drawn to within tolerance (as a minimum, examiners should check intersections with axes, candidates do not need to mark these).
	B1	For two of the lines drawn to within tolerance (as a minimum, examiners should check intersections with axes, candidates do not need to mark these).
	B1	For all three lines correctly drawn to within tolerance (as a minimum, examiners should check intersections with axes, candidates do not need to mark these).
As a minimum, lines must intersect with other for marks in (a) and (b) to be awarded		
(b)	**B1ft**	For the correct **enclosed** region shaded in or out or for R clearly labelled. The **ft** mark can only be awarded if 3 distinct lines have been drawn and it's clear they've shaded on the correct 'side' for each of their lines. If there's no labelling and it's not clear which line is which, this mark cannot be awarded.
Part c of this question states "using your graph"…… Therefore solutions which obviously use exact coordinates of intersection points having used a graphical calculator or from working algebraically can only score M0 M1 A0.		
(c)	**M1**	For reading **from the graph** at least one point of intersection using their lines. The pair used for this and the next method mark must be within the tolerance of ± 0.1 of the values shown in the table. Any solutions which work out the values algebraically will not gain this mark or the final accuracy mark, but may gain the next method mark. Occasionally, students are working out the non-integer coordinates from algebra or from a calculator but reading (-4, -3) from the graph and using this to find the value of P. In this case, we can apply bod (benefit of the doubt) and this mark can be awarded if subbed in to find the value of P.
	M1	For a correct substitution to find the value of P from at least one set of their coordinates of the point of intersection. This is not a dependent mark, so they can use any one of their pairs of values even if it doesn't fall in tolerance.
	A1	For the correct least value of -28.2 Allow a value between -28.9 and -27.5 so long as this follows through from their values. Do not allow a value out of range to be rounded to a value within range.

Type MS · Shading or describing a region

Answer 1 of 11 · Question 5 part b · 1 mark

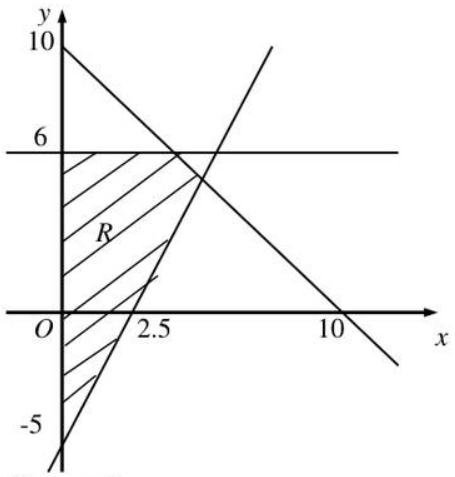
2023 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Shading or describing a region

Answer 2 of 11 · Question 2 part b · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number		
2(a)		B1 B1 B1 (3)
(b)	Correct shading (in or out)	B1 (1) [4]

Part	Mark	Notes
(a)	B1	For any one correct line from $y=6$, $y+x=10$, $y=2x-5$ Line / y intercept / x intercept $y=6$ / 6 / No intercept $y+x=10$ / 10 / 10 $y=2x-5$ / –5 / 2.5 Accept unambiguous indication on labelled axes. **Note:** • The line must cross both axes for the award of a mark • Accept an unruled line provided the intention is clear. Look for the intersections on the axes.
	B1	For any two correct lines from $y=6$, $y+x=10$, $y=2x-5$
	B1	All three correct lines $y=6$, $y+x=10$, $y=2x-5$
	B1	For the correct region shaded in or out. R does not need to be written onto the sketch.

Type MS · Shading or describing a region

Answer 2 of 11 · Question 2 part b · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Shading or describing a region

Answer 3 of 11 · Question 4 part b · 1 mark

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

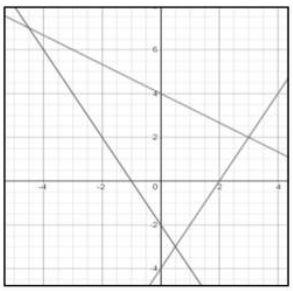
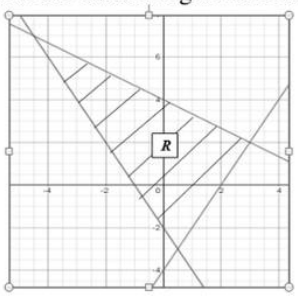
		$1-\frac{}{4}$ $\frac{'5'}{12} \neq \frac{5}{16}$, $\frac{'5'}{12} \neq \frac{135}{1024}$
	A1	For the correct value of $S = \frac{5}{3}$ **Note:** Must be the exact value.

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Shading or describing a region

Answer 3 of 11 · Question 4 part b · 1 mark

2022 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
4 (a)	$y = 2x - 4$ Intersections with axes at $(0, -4)$ $(2, 0)$ $2x + 3y = 12$ Intersections with y axes at $(0, 4)$ and. $(6, 0)$ $y + 2x + 2 = 0$ Intersections with y axes at $(0, -2)$ and. $(-1, 0)$ 	B1 B1 B1 [3]
(b)	For the correct region shaded in or out 	B1ft [1]
(c)	Points of intersection are: $(0.5, -3)$ $(3, 2)$ $(-4.5, 7)$ Vertex: $(0.5, -3)$ $(3, 2)$ $(-4.5, 7)$ $P = x - 2y$: 6.5 −1 −18.5 Least For $P = -18.5$	M1 A1 dM1 A1 [4]
	ALT- objective line approach	

Type MS · Shading or describing a region

Answer 3 of 11 · Question 4 part b · 1 mark

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

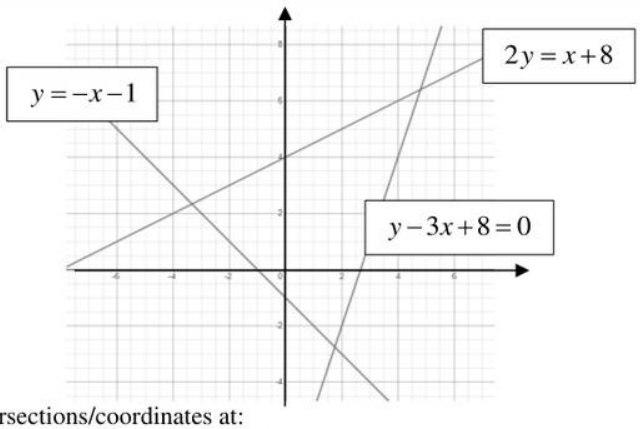
	Slope of objective line is $\frac{1}{2}$ $\left(-\frac{9}{2}, 7\right)$ $P = -\frac{9}{2} - 2(7)$ For $P = -18.5$	[M1 A1 M1 A1]
		Total 8 marks

Type MS · Shading or describing a region

Answer 4 of 11 · Question 4 part b · 1 mark

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

4(a)



Intersections/coordinates at:

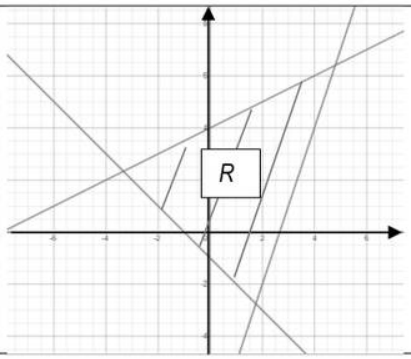
$y = -x - 1 \Rightarrow (-1, 0)$ and $(0, -1)$

$2y = x + 8 \Rightarrow (-2, 3)$ and $(0, 4)$

$y - 3x + 8 = 0 \Rightarrow (3, 1)$ and $(2, -2)$

B3
[3]

(b)



B1ft
[1]

(c)

Intersection	$\left(-3\frac{1}{3}, 2\frac{1}{3}\right)$	$\left(1\frac{3}{4}, -2\frac{3}{4}\right)$	$\left(4\frac{4}{5}, 6\frac{2}{5}\right)$
P	$\frac{44}{3}$	-10.75	-1.6

M1A1

$P_{max} = \frac{44}{3}$

$P_{min} = -10.75$

dM1A1
[4]

Type MS · Shading or describing a region

Answer 4 of 11 · Question 4 part b · 1 mark

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Total 8 marks

Part	Mark	Notes
(a)	B1	For at least one line drawn correctly. Please check their intersections/coordinates carefully, these lines are to be drawn accurately.
	B1	For at least two lines drawn correctly
	B1	For all three lines drawn correctly.
(b)	B1ft	For an enclosed region shaded in or out. Ft their lines from (a) provided the region is closed. *R* does not need to be written in the region.
(c)	M1	For an attempt to find the coordinates of at least one point of intersection either by simultaneous equations or by reading off from their graph. The question does not specify 'using your graph' so either method is fine.
	A1	For at least one set of correct coordinates. If the candidate uses the graph, then allow all values awrt: $(-3.3\pm0.1,\ 2.3\pm0.1)$ for $\left(-3\frac{1}{3}, 2\frac{1}{3}\right)$ $(1.7\pm0.1,\ -2.7\pm0.1)$ for $\left(1\frac{3}{4}, -2\frac{3}{4}\right)$ $(4.8\pm0.1,\ 6.4\pm0.1)$ for $\left(4\frac{4}{5}, 6\frac{2}{5}\right)$
	dM1	For using at least one set **their** coordinates of their intersections to find any value of *P* Explicit substitution need not be seen provided it is clear candidates are using the expression for *P*. Accept awrt: 14.6 ± 0.1 or -10.7 ± 0.1 or -1.6 ± 0.1 **Note** this is dependent on the previous M mark If their values are incorrect and you see no working this is M0 If their values are correct, you see no working and no value of *P* is correct M0
	A1	For both the correct minimum and maximum values. $P_{max} = 14.66$ accept awrt 14.7 ± 0.3 $P_{min} = -10.75$ accept awrt -10.7 ± 0.3

Type MS · Shading or describing a region

Answer 4 of 11 · Question 4 part b · 1 mark

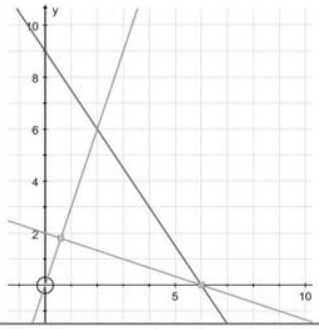
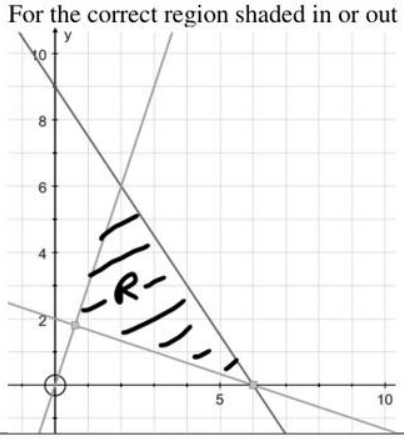
2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Shading or describing a region

Answer 5 of 11 · Question 4 part b · 1 mark

2025 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

4 (a)	$3x+2y=18$ Intersections with axes at (0,9) (6,0) $x+3y-6=0$ Intersections with axes at (0,2) (6,0) $y=3x$ Coordinates at (0,0) (2, 6) 	B1 B1 B1 [3]
(b)	For the correct region shaded in or out 	B1ft [1]
(c)	Points of intersection are: (0.6,1.8) (2,6) (6,0) Vertex | (0.6,1.8) | (2,6) | (6,0) $P=2x-y$ | −0.6 | −2 | 12 Least For $P=-2$	M1 A1 dM1 A1 [4]
	ALT – objective line approach	
	Slope of objective line is 2 (2,6) $P=2\times2-6=-2$ For $P=-2$	[M1 A1 dM1 A1]
		Total 8 marks

Type MS · Shading or describing a region

Answer 5 of 11 · Question 4 part b · 1 mark

2025 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1B1B1	B1 for one line correct, B1B1 for two lines correct, B1B1B1 for all three lines correct: $3x+2y=18$ Intersections with axes at (0,9) (6,0) $x+3y-6=0$ Intersections with axes at (0,2) (6,0) $y=3x$ Intersections with axes at (0,0) (2, 6)
(b)	B1ft	For the correct region marked – allow shaded in or out. Ft for shading the **closed** region from their three lines.
(c)	M1	For attempting to find at least **one** coordinate pair of their intersections: Either by reading values from their graph (might be written on the graph), **or** by solving simultaneous equations leading x=… y=....
	A1	Finds all **three** correct points of intersection: Allow point $(0.6,1.8)$ to be $(0.6\pm0.1,1.8\pm0.1)$ (0.6,1.8) | (2,6) | (6,0)
	dM1	For substituting any **one** point of intersection (could be from reading from the graph or solved by simultaneous equations) into the given P ft their coordinates. Any correct P value: $P=-2$, $P=-0.6$ or $P=12$ implies this mark. Dependent on the previous M mark
	A1	For $P=-2$ from testing at least **two** points, **and** all **three** points of intersection must be correct $(0.6,1.8)$, $(2,6)$ and $(6,0)$
ALT – objective line approach		
	M1	For an attempt to use objective line approach. Identified that the slope of the objective line is 2. Identified that the intersection of $3x+2y=18$ and $y=3x$ is where P is the least.
	A1	For finding the correct coordinates (2,6)
	dM1	For substituting their (2,6) into P.
	A1	For identifying $P=-2$

Type MS · Shading or describing a region

Answer 5 of 11 · Question 4 part b · 1 mark

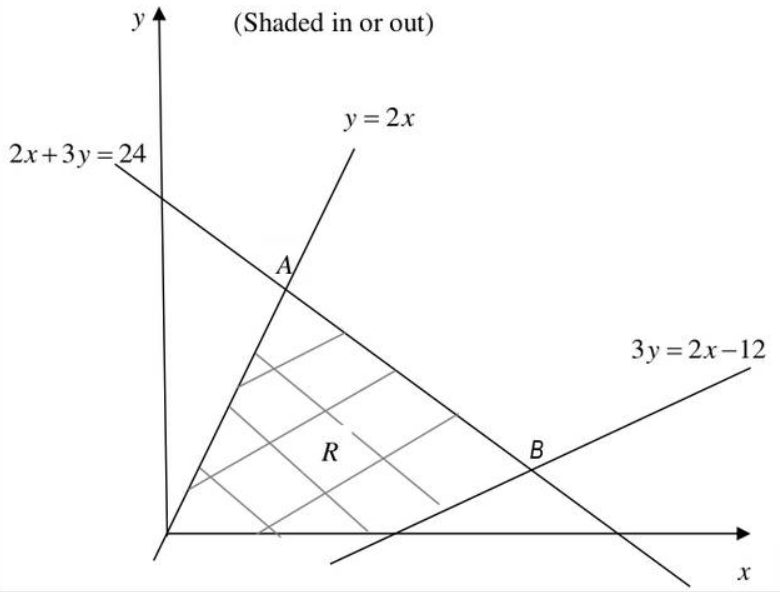
2025 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Shading or describing a region

Answer 6 of 11 · Question 5 part b · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

5 (a)	$2x+3y=24,\quad y=2x,\quad 3y=2x-12$	M1 A1A1 [3]
(b)		B1 [1]
(c)	Greatest value of F is at A, intersection of lines $y=2x$ and $2x+3y=24$ Greatest value of F is at (3, 6) Greatest value of $F = 2\times3+5\times6=36$	M1 A1 A1ft [3]
		Total 7 marks

(a)	M1	Attempt to find two points on any one of the straight lines. Look for two correct points (stated, plotted or implied by a correct line), or for any incorrect point from working which shows an appropriate substitution of one variable with an attempt to solve for the second variable.
	A1	For any two correct lines.
	A1	For three correct lines.
(b)	B1	Correct region, identified by shading in or outside the region. Label R is not required but it needs to be correct if it is shown.
(c)	M1	Identify A as the point with coordinates that give the greatest value of F, or find the value of F at A and B, ft their graph, or draw line $2x+5y=k$ for at least one value of k.
	A1	Correct coordinates (3, 6) for the point identified to give the greatest value of F. May be read from graph or implied by working.
	A1ft	(F =) 36 ft their coordinates for A. Note that this mark is shown as B1 on ePEN.

Type MS · Shading or describing a region

Answer 6 of 11 · Question 5 part b · 1 mark

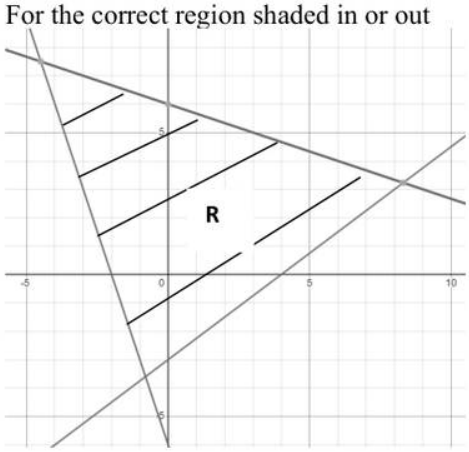
2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Shading or describing a region

Answer 7 of 11 · Question 5 part b · 1 mark

2025 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5(a) (i) (ii) (iii)	For each line drawn correctly $3x-4y=12$ Intersections with axes at (0, -3) and (4, 0) $y+6+3x=0$ Intersections with axes at (0, -6) and. (-2, 0) $3y=18-x$ Coordinates at (0, 6) (3, 5) (6, 4)	 B1 B1 B1 [3]
(b)	For the correct region shaded in or out 	B1ft [1]
(c)	For attempting to **read off** the correct coordinates of at least one intersection. Vertex: $(-0.8,-3.6)$ $(8.3,\ 3.2)$ $(-4.5,\ 7.5)$	M1
	$P=3x-2y$: 4.8 | 18.5 | −28.5 Greatest | Least	dM1 A1caoA1cao [4]
		Total 8 marks

Type MS · Shading or describing a region

Answer 7 of 11 · Question 5 part b · 1 mark

2025 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Part</th><th>Mark</th><th>Notes</th></tr>
<tr><td>5 (a)</td><td>B3</td><td>Award B1 for each line drawn correctly.</td></tr>
<tr><td>(b)</td><td>B1ft</td><td>For the correct region shaded in or out.
You do not need to see R marked. Follow through with candidate's lines, where a closed region has been created.</td></tr>
<tr><td rowspan="4">(c)</td><td>M1</td><td>For at least one set of coordinates read off from their graph.
If fractional values such as -4/5. -18/5, 108/13 etc award M0.</td></tr>
<tr><td>dM1</td><td>For finding at least one value for P using their coordinates.</td></tr>
<tr><td>A1cao</td><td>For either $P_{\text{greatest}} = 18.5$ or $P_{\text{least}} = -28.5$</td></tr>
<tr><td>A1cao</td><td>For both $P_{\text{greatest}} = 18.5$ and $P_{\text{least}} = -28.5$
Mark should only be awarded for fully correct work with all three values correct.</td></tr>
<tr><td></td><td></td><td>Note on tolerances
For each point on the graph, we accept candidate values within "half a square" of the correct value.
Thus, for the first vertex, we accept x values in (-0.9 - -0.7) and corresponding y values within (-3.7 to -3.5)

Co-ordinates:

<table>
<tr><td>Vertices</td><td>Accept awfw</td><td></td><td>P awfw</td></tr>
<tr><td>(-0.8, -3.6)</td><td>(-0.9, -0.7)</td><td>(-3.7 -3.5)</td><td>(4.3, 5.3)</td></tr>
<tr><td>(8.3, 3.2)</td><td>(8.2, 8.4)</td><td>(3.1, 3.3)</td><td>(18.0, 19.0)</td></tr>
<tr><td>(-4.5, 7.5)</td><td>(-4.6, -4.4)</td><td>(7.4, 7.6)</td><td>(-29.0, -28.0)</td></tr>
</table>
</td></tr>
</table>

Type MS · Shading or describing a region

Answer 7 of 11 · Question 5 part b · 1 mark

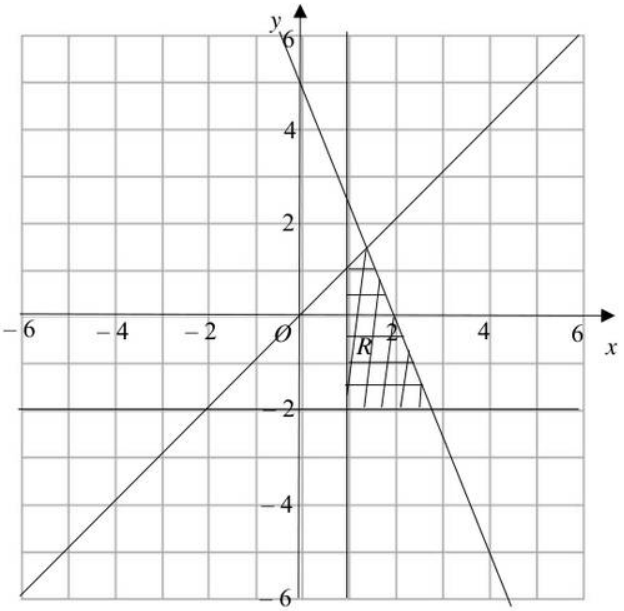
2025 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Shading or describing a region

Answer 8 of 11 · Question 2 part b · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

2 (a)		B1 B1 (2)
(b)		B1 B1 (2) **[4]**

Part	Mark	Additional Guidance
(a)	B1 (i)	For **either** correct line drawn The correct intersection on the axis for $5x+2y=10$ are $(2,\ 0)$ and $(0,\ 5)$ Coordinates for the $y=x$ are $(0,\ 0)$ $(1,\ 1)$ $(2,\ 2)$ $(3,\ 3)$ etc
	B1 (ii)	For **both** lines drawn correctly
(b)	B1	For both lines $y=-2$ and $x=1$ drawn correctly
	B1	The correct region shaded in or out You do not need to see the label R

Type MS · Shading or describing a region

Answer 8 of 11 · Question 2 part b · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Shading or describing a region

Answer 9 of 11 · Question 1 part b · 2 marks

2024 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Shading or describing a region

Answer 9 of 11 · Question 1 part b · 2 marks

2024 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

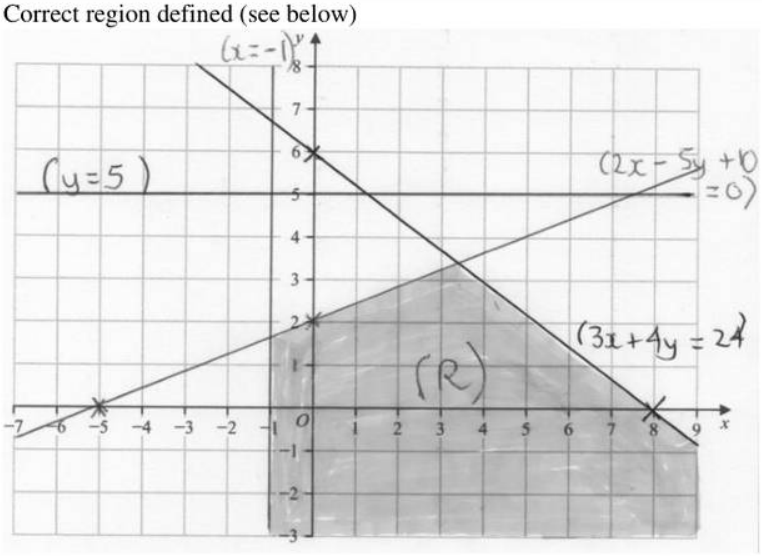
Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

Type MS · Shading or describing a region

Answer 9 of 11 · Question 1 part b · 2 marks

2024 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
1 (a)(i)	$3x+4y=24$ drawn	B1 [1]
(a)(ii)	$2x-5y+10=0$ drawn	B1 [1]
(b)	$x=-1$ drawn	M1 (B1 on ePen)
	Correct region defined (see below) 	A1ft (B1 on ePen) [2]
		Total 4 marks

Type MS · Shading or describing a region

Answer 10 of 11 · Question 1 part b · 2 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication

from the substitution of correct values and then proceeding to a solution.

Type MS · Shading or describing a region

Answer 10 of 11 · Question 1 part b · 2 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

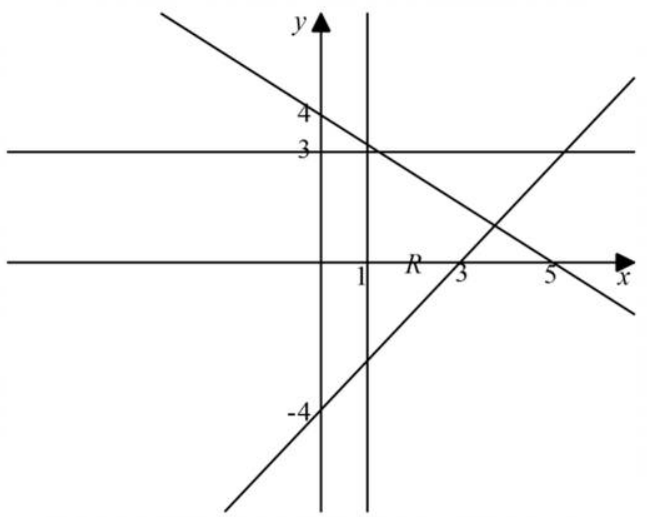
Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Type MS · Shading or describing a region

Answer 10 of 11 · Question 1 part b · 2 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
1 (a)(i)	$4x+5y=20$ drawn	B1 (1)
(a)(ii)	$3y-4x=-12$ drawn	B1 (1)
(b)	$y=3$ and $x=1$ drawn	B1
	Correct region defined (see below) 	B1ft (2)
		Total 4 marks

Type MS · Shading or describing a region

Answer 11 of 11 · Question 2 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2 (a)	Line l_1 $m = \frac{-8}{4} = -2$ $y - 8 = -2(x) \Rightarrow y + 2x = 8$ Line l_2 $m = \frac{-4}{6} = -\frac{2}{3}$ $y - 4 = -\frac{2}{3}(x) \Rightarrow 3y + 2x = 12$	M1A1 A1 [3]
(b)	$x \geq 0$ $y + 2x \leq 8$ $3y + 2x \geq 12$ Accept < and >	B1B1ftB1ft [3]
		Total 6 marks

Type MS · Shading or describing a region

Answer 11 of 11 · Question 2 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks